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Research Article

An Automatic Ionogram Scaling Method Based on Temporal Continuity Modeling

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Abstract

Ground-based vertical ionospheric sounding is an essential observational technique for obtaining information on ionospheric structure. Efficient and accurate automatic ionogram scaling is of great significance for practical applications. In recent years, deep learning methods have been gradually applied to automatic ionogram scaling tasks and have shown considerable potential in layer trace extraction, noise suppression, and critical parameter retrieval. However, most existing deep learning methods treat this task as a single-frame static segmentation problem, relying solely on information from the current frame and struggling to exploit the continuous evolutionary information between consecutive ionograms. As a result, the scaling accuracy still has room for improvement under complex ionogram conditions. To address this issue, this paper proposes TCAS-Net, an automatic

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current frame. It employs a shared encoder to extract spatial features from multiple time steps and introduces ConvLSTM in the bottleneck feature space for explicit temporal fusion, thereby providing supplementary temporal prior information for trace identification in the current frame. Experimental results show that TCAS-Net achieves a mean absolute error of 5.6300 km for the minimum virtual height and 0.0754 MHz for the maximum critical frequency, demonstrating that explicitly incorporating the continuous evolutionary relationship between consecutive ionograms into the automatic scaling process of the current frame helps improve the model's scaling performance.

Plain Language Summary

Efficient and accurate automatic scaling of ionograms is of great significance for ionospheric detection. However, most existing deep learning methods only use single-frame information, making it difficult to extract the continuous evolution relationship between consecutive ionograms. To address this, this paper proposes TCAS-Net, a deep learning network based on temporal continuity modeling. The network takes consecutive ionogram frames as input and introduces a temporal fusion module to capture the evolutionary information between adjacent observations. Experimental results show that TCAS-Net achieves mean absolute errors of 5.6300 km for the minimum virtual height and 0.0754 MHz for the maximum critical frequency. Compared with existing methods, TCAS-Net demonstrates improved accuracy in key parameter extraction and better overall stability in automatic ionogram scaling.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The ionogram data set used in this study is available at the website **http://www.geophys.ac.cn/ArticleDataInfo.asp?MetaId=205** and we are grateful to the creator (Xiao, **2020**). The source code of TCAS-Net has been deposited in Zenodo and is publicly available at **https://doi.org/10.5281/zenodo.20195057** (X. Chen & Liu, **2026**). The Zenodo repository includes the data

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parameter extraction code for calculating D-MEH and D-MCF. The temporal samples are constructed in the original chronological order of the ionograms, and the final model setting uses three consecutive frames as input with current-frame supervision. The deep learning models were built using the TensorFlow library (Abadi et al., **2015**) (**https://www.tensorflow.org**), Keras library (Chollet, **2015**) (**https://keras.io**) and the Segmentation Models library (Yakubovskiy, **2019**) (**https://github.com/qubvel/segmentation_models**). All figures were made with the free and open software Matplotlib (Caswell et al., **2020**; Hunter, **2007**) (**https://matplotlib.org/**).

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